Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Density of some aq. solution (60% by wt.) is lowered upto $1.2\mathrm{gm/cc}$ by addition of water. If initial density was $1.6\mathrm{gm/cc}$ then new approximate percentage by wt. will be (Take density of water $= 1\mathrm{gm/cc}$)

(A) 45%

(B) 39%

(C) 27%

(D) 19%

Q.2

At $20^{\circ}\mathrm{C}$ and 1.00 atm partial pressure of hydrogen, $18\ \mathrm{mL}$ of hydrogen measured at STP, dissolves in $1\ \mathrm{L}$ of water. If water at $20^{\circ}\mathrm{C}$ is exposed to a gaseous mixture having a total pressure of 1400 Torr (excluding the vapour pressure of water) and containing $68.5\%\mathrm{H_2}$ by volume, what volume of $\mathrm{H_2}$, measured at STP, will dissolve in $1\mathrm{L}$ of water ?

(A) $18\ \mathrm{mL}$

(B) $12\ \mathrm{mL}$

(C) $23\ \mathrm{mL}$

(D) $121\ \mathrm{mL}$

Q.3

The elements A and B form compounds having molecular formulae $\mathrm{AB_2}$ and $\mathrm{AB_4}$, when dissolved in $20\ \mathrm{g}$ of $\mathrm{C_6H_6}$. $\mathrm{lg}\ \mathrm{AB_2}$ lowers the freezing point by $2.3\ \mathrm{K}$ whereas $1.0\ \mathrm{g}$ of $\mathrm{AB_4}$ lowers it by $1.3\ \mathrm{K}$. The molal depression constant for benzene is $5.1\ \mathrm{K\ kg\ mol^{-1}}$. The atomic masses of A and B are, respectively -

(A) $\mathrm{A} = 26,\ \mathrm{B} = 42.64$

(B) $A = 31.72, B = 47.02$

(C) $\mathrm{A} = 13.11,\ \mathrm{B} = 24.25$

(D) $\mathrm{A} = 19.17,\ \mathrm{B} = 3.01$

Q.4

π_1, π_2, π_3 and π_4 are the osmotic pressure of 5% (mass / volume) aqueous solution of each of urea , fructose , sucrose and $\mathrm{KCN}\ (\alpha = 1)$ respectively at certain temperature The correct order of their magnitudes is

(A) $\pi_1 > \pi_4 > \pi_2 > \pi_3$

(B) $\pi_1 < \pi_4 < \pi_2 < \pi_3$

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(C) $\pi_4 > \pi_1 > \pi_2 > \pi_3$

(D) $\pi_4 > \pi_1 > \pi_3 > \pi_2$

Q.5

A mixture of chlorobenzene and water (which are virtually immiscible) boils at $90.3^{\circ}\mathrm{C}$ at an external pressure of $740.2\ \mathrm{mm}$. The vapour pressure of pure water at $90.3^{\circ}\mathrm{C}$ is $530.1\ \mathrm{mm}$. What is the percent weight of chlorobenzene in the distillate ?

(A) 20.21

(B) 37.19

(C) 28.74

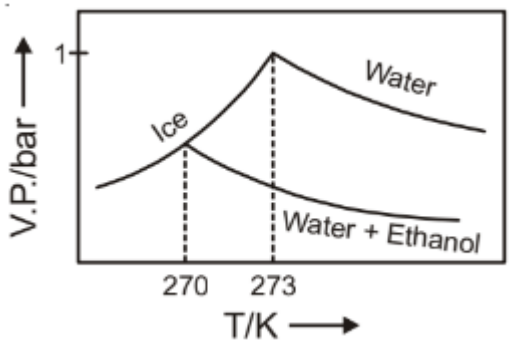
(D) 71.26

Q.6

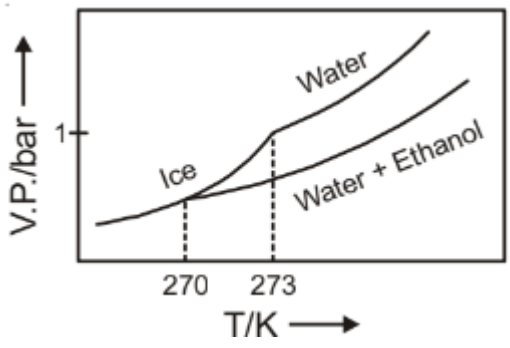
Pure water freezes at $273\ \mathrm{K}$ and $1\mathrm{bar}$. The addition of $34.5\ \mathrm{g}$ of ethanol to $500\ \mathrm{g}$ of water changes the freezing point of the solution. Use the freezing point depression constant of water as $2\ \mathrm{K\ kg\ mol^{-1}}$. The figures shown below represent plots of vapour pressure (V.P.) versus temperature (T). [Molecular weight of ethanol is $46\ \mathrm{g\ mol^{-1}}$]

Among the following, the option representing change in the freezing point is

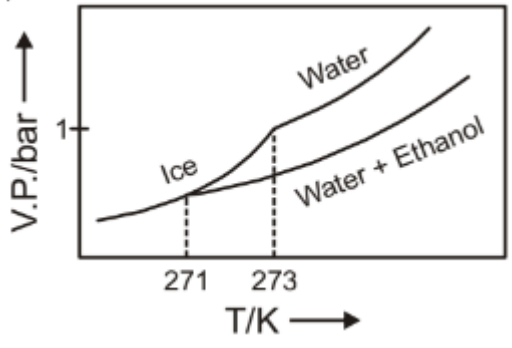
(A)



(B)

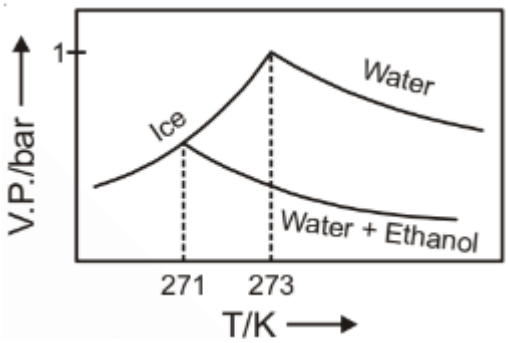


(C)



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(D)



Q.7

A gaseous mixture of helium and oxygen is found to have a density of $0.5\ \text{gdm}^{-3}$ at 25°C and 0.662 atm. What is the percentage by mass of helium in this mixture?
(Given: $\text{R} = 0.08\ \text{L atm mol}^{-1}\ \text{K}^{-1}$, $\frac{5960}{331} = 18$)

(A) 11.11

(B) 14.23

(C) 23.18

(D) 18.13

Q.8

The vapour pressure of a pure liquid A is $70\ \text{mm}$ of Hg at $300\ \text{K}$. It forms an ideal solution with an another liquid B. The mole fraction of B in the solution is 0.2 and total pressure of the solution is $84\ \text{mL}$ of Hg at $300\ \text{K}$. The vapour pressure of pure liquid B at $300\ \text{K}$ would be -

(A) $140\ \text{mm}$ of Hg

(B) $70\ \text{mm}$ of Hg

(C) $35\ \text{mm}$ of Hg

(D) $280\ \text{mm}$ of Hg

Q.9

Mark out the correct statement (s) -

(A) For a binary solution AB, if $\text{F}_{\text{A-B}} > \text{F}_{\text{A-A}}$ and $\text{F}_{\text{B-B}}$, there is negative deviation from ideality or Rault's law.

(B) If $\text{F}_{\text{A-B}} = \text{F}_{\text{A-A}}$ and $\text{F}_{\text{B-B}}$, binary solution AB formation is not spontaneous as $\Delta\text{H}_{\text{mixing}} = 0$.

(C) If $\text{F}_{\text{A-B}} < \text{F}_{\text{A-A}}$ and $\text{F}_{\text{B-B}}$, $\Delta\text{H}_{\text{mixing}} > 0$ and solution formation is spontaneous.

(D) There is deviation from Raoult's law in every case if $\Delta\text{H}_{\text{mixing}} \neq 0$

Q.10

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The total vapour pressure of a binary solution is given by $p = (110x_A + 125x_B)\,\text{mmHg}$, where x_A and x_B are the mole fractions of components A and B, respectively. It suggests that -

(A) the vapour pressure of solution $(\text{n}_\text{A} = \text{n}_\text{B})$ is less than the pure B component.

(B) the vapour pressure of solution $(\text{n}_\text{A} = \text{n}_\text{B})$ is more than that of pure component A

(C) the vapour pressure of pure A component is $110\ \text{mmHg}$ and that of pure B is $125\ \text{mmHg}$.

(D) the vapour pressure of pure A and B are $125\ \text{mmHg}$ and $110\ \text{mmHg}$, respectively.

Q.11

Two liquids A and B form an ideal solution. The solution has a vapour pressure of 700 torr at 80°C. It is distilled till $2/3\text{rd}$ of the solution is collected as condensate. The composition of the condensate is $\text{x}'_\text{A} = 0.75$ and that of the residue is $\text{x}''_\text{A} = 0.30$. If the vapour pressure of the residue at 80°C is 600 torr, which of the following is/are true -

(A) The composition of the original liquid was $\text{x}_\text{A} = 0.6$

(B) The composition of the original liquid was $x_A = 0.4$

(C) $\text{P}^\circ_\text{A} = \frac{2500}{3}$ Torr

(D) $\text{P}^\circ_\text{B} = 500$ Torr

Q.12

Paragraph:
The total vapour pressure of a binary solution is given by $p = (110x_A + 125x_B)\,mm\text{Hg}$ where x_A and x_B are the mole fractions of components A and B, respectively.
Question:
The vapour pressure of pure liquid component A is -

(A) 135

(B) 110

(C) 125

(D) 15

Q.13

Paragraph:
The total vapour pressure of a binary solution is given by $p = (110x_A + 125x_B)\,mm\text{Hg}$ where x_A and x_B are the mole fractions of components A and B, respectively.
Question:
The mole fraction of A in the solution is 0.3. The mole fraction of B in the vapour phase, y_B, is -

(A) 0.36

(B) 0.73

(C) 0.27

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(D) 0.46

Q.14

Paragraph:

The total vapour pressure of a binary solution is given by $p = (110x_A + 125x_B)\, mm\text{Hg}$ where x_A and x_B are the mole fractions of components A and B, respectively.

Question:

For the equimolar solution of A and B, which of the following is a correct statement?

(A) $\mathrm{p} > \mathrm{p_A}^\circ$

(B) $\mathrm{p} < \mathrm{p_A^\circ}$

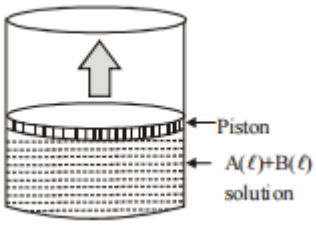
(C) $p = {p_A}^\circ$

(D) None of these

Q.15

Paragraph:

An ideal solution of two volatile liquids $\mathrm{A}(\ell)\&\ \mathrm{B}(\ell)$ is vaporized by raising the piston slowly at constant temperature. Initially in liquid mixture mole $\%$ of A was 60% at $1\ \text{atm}$. Vapour pressures of pure liquids A \& B are 100 and $400\ \text{mm}$ Hg respectively.



Question:

What is mole fraction of A in solution at when vapour pressure of solution is 0.5 atm ?

(A) $2/3$

(B) $3/5$

(C) $1/15$

(D) $4/15$

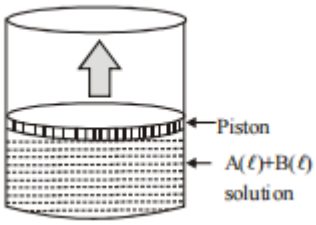
Q.16

Paragraph:

An ideal solution of two volatile liquids $\mathrm{A}(\ell)\&\ \mathrm{B}(\ell)$ is vaporized by raising the piston slowly at constant temperature. Initially in liquid mixture mole $\%$ of A was 60% at $1\ \text{atm}$. Vapour pressures of pure liquids A \& B

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are 100 and $400\ mm$ Hg respectively.



Question:

What is vapour pressure of solution when last drop of liquid mixture is left?

(A) $1/7$

(B) $2/3$

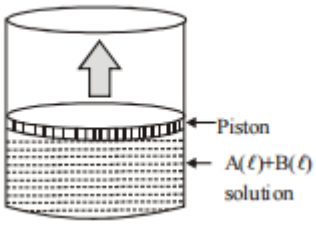
(C) $6/7$

(D) $1/3$

Q.17

Paragraph:

An ideal solution of two volatile liquids $A(\ell)\&\ B(\ell)$ is vaporized by raising the piston slowly at constant temperature. Initially in liquid mixture mole $\%$ of A was 60% at $1\ atm$. Vapour pressures of pure liquids A \& B are 100 and $400\ mm$ Hg respectively.



Question:

As piston is lifted which of options is correct -

(A) Mole fraction of A in liquid solution decreases

(B) Mole fraction of B in liquid solution increases

(C) Mole fraction of A in vapour phase decreases

(D) None of these

Q.18

Match the column:

Column I (Solution)	**Column II (Characteristics)**
(A) $C_2H_5OH - H_2O$	(p) $\Delta S > 0$
(B) $H_2O - HNO_3$	(q) $\Delta V > 0$
(C) $CH_3OH - CH_3CH_2OH$	(r) $\Delta H < 0$
(D) $CCl_4 - CH_3OH$	(s) $p_1 > p_1^0 x_1$

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(A) A-p,q,s; B-p,r ; C- p ; D-p,q,s;

(B) A-p,q; B-r ; C- p,q,s ; D-p,q,s;

(C) A-p,q,s; B-p,r ; C- p,q,r,s ; D-p,q,s;

(D) A-r,s; B-p,r ; C- p ; D-p,q,s;

Q.19

An aqueous solution of an acid is so weak that it can be assumed to be practically un-ionised when boiled at $100.4^{\circ}\mathrm{C}, 25\ \mathrm{mL}$ of this solution was neutralised by $38.5\ \mathrm{mL}$ of $1\ \mathrm{N}$ solution of NaOH. Choose the basicity of the acid if $\mathrm{K_b}$ for water is $0.52\ \mathrm{K\ mol^{-1}\ kg}$.

Q.20

A certain mass of a substance when dissolved in $100\ \mathrm{g} \mathrm{C_6H_6}$, lowers the freezing point by $1.28^{\circ}\mathrm{C}$. The same mass of solute dissolved in $100\ \mathrm{g}$ water lowers the freezing point by $1.40^{\circ}\mathrm{C}$. If the substance has normal molecular weight in benzene and is completely ionised in water, how many ions does it dissociate in water?
$\left[\mathrm{K_f\ (H_2O)} = 1.86\ \mathrm{K\ mol^{-1}\ kg}\ \text{and}\ \mathrm{K_f\ (C_6H_6)} = 5.12\ \mathrm{K\ mol^{-1}\ kg}\right]$

Answers & Solutions

Q.1 Answer:

27%

Solution:

mass of 1 litre of solution $= 1600\mathrm{gms.}$
mass of solute $= 960\mathrm{gm}$
Let V **litre of water is added.**
new vol $= (1+\mathrm{V})$ **litre** $= (1+\mathrm{v})1000\mathrm{cc}$
new mass $= 1600 + \mathrm{V}(1000)\mathrm{gms}$
$1.2 = \frac{1600+1000\ \mathrm{V}}{(1+\mathrm{V})1000} = \frac{1.6+\mathrm{V}}{1+\mathrm{V}}$
$1.6 + \mathrm{V} = 1.2 + 1.2\ \mathrm{V}$
$0.2\ \mathrm{V} = 0.4$
$\mathrm{V} = 2$ **litre**
new wt. $\% = \frac{960\times100}{1600+2000} = 27\%$ **approx**

Q.2 Answer:

$23\ \mathrm{mL}$

Solution:

According to Henry's law, $\frac{\mathrm{V_2}}{\mathrm{V_1}} = \frac{\mathrm{p_2}}{\mathrm{p_1}}$
or $\frac{\mathrm{V_2}}{18\ \mathrm{mL}} = \frac{1.26}{1.00}$ **or** $\mathrm{V_2} = 23\ \mathrm{mL}$

Q.3 Answer:

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$\mathrm{A} = 26,\ \mathrm{B} = 42.64$

Solution:

Suppose atomic masses of A and B are x and y, respectively.

Molar mass of $\mathrm{AB_2} = \mathrm{x} + 2\mathrm{y}$

$\lg \mathrm{AB_2} = \frac{1}{\mathrm{x}+2\mathrm{y}} \mathrm{mol}$

Molar mass of $\mathrm{AB_4} = \mathrm{x} + 4\mathrm{y}$

$\lg \mathrm{AB_4} = \frac{1}{x+4y}\ \mathrm{mol}$

$\mathrm{m}\left(\mathrm{AB_2}\right) = \frac{1/(\mathrm{x}+2\mathrm{y})}{20} \times 1000 = \frac{50}{\mathrm{x}+2\mathrm{y}}$

$\Delta \mathrm{T_f} = \mathrm{K_f} \times \mathrm{m}$

$2.3 = 5.1 \times \frac{50}{\mathrm{x}+2\mathrm{y}}$

or $\quad x + 2y = 110.87$

Similarly, for the 2nd **case**

$x + 4y = 196.15$

(2)

Solving eqs. (1) and (2), we get, $x = 25.59, y = 42.64$

Q.4 Answer:

$\pi_4 > \pi_1 > \pi_2 > \pi_3$

Solution:

Calculate molarity for each solution

$$\text{molarity of urea solutioon} = \frac{\left(P\frac{W}{V}\right) \times 10}{GMW} = \frac{5 \times 10}{60} = 0.83$$

$$\text{Molarity of fructose solution} = \frac{\left(P\frac{W}{V}\right) \times 10}{GMW} = \frac{5 \times 10}{180} = 0.27$$

$$\text{Molarity of sucrose solution} = \frac{\left(P\frac{W}{V}\right) \times 10}{GMW} = .\frac{5 \times 10}{342} = 0.14$$

Q.5 Answer:

71.26

Solution:

At boiling point $\mathrm{P'_M} = 740.2\ \mathrm{mm}$

$\mathrm{P'_{H_2O}} = 530.1\ \mathrm{mm}$

$\therefore \mathrm{P'_{chlorobenzene}} = 740.2 - 530.1$

$= 210.1\ \mathrm{mm}$

Also, $\mathrm{P'} = \mathrm{P_M} \times$ **mole fraction**

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$530.1 = 740.2\times$ **mole fraction of** H_2O

$\therefore$ **mole fraction of** $H_2O = 0.716$

$\therefore$ **mole fraction of** $C_6H_5Cl = 0.284$

Let a g_2O **and b g chlorobenzene be present in distillate or** $\frac{w_{H_2O}/18}{\frac{w_{CB}}{112.5}+\frac{w_{H_2O}}{18}} = 0.716$....**(i) and** $\frac{w_{H_2O}/112.5}{\frac{w_{CB}}{112.5}+\frac{w_{H_2O}}{18}} = 0.284$.

By eq. (i)/(ii)

or $\frac{a}{18} \times \frac{112.5}{b} = \frac{0.716}{0.284}$

or $\frac{a}{b} = \frac{12.89}{31.5} = 0.403$

Let total distillate be 100 g

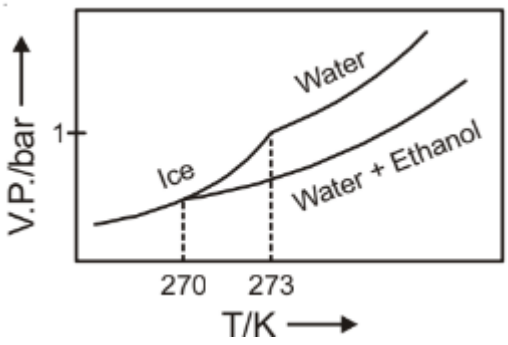
$\therefore a + b = 100 \ldots . (iv)$

By eqs. (iii) and (iv)

$b = 71.27\%$

$a = 28.73\%$

Q.6 Answer:



Solution:

$$\Delta T_f = iK_f \left[\frac{W_2 \times 1000}{M_2 \times W_1}\right]$$

$$= 1 \times 2 \left[\frac{34.5 \times 1000}{46 \times 500}\right]$$

$$= 3 \text{ K}$$

$$273(K) - T_f = 3(K)$$

$$\Rightarrow \quad T_f = 270 \text{ K}$$

Also, with decrease in temperature, V.P. decreases.

$\therefore$ **Graph** (B) **is correct.**

Q.7 Answer:

11.11

Solution:

Let the mean molar mass of the mixture be $= M$.

From ideal gas law,

we know that, $PM = dRT$

Where, d **= density of the gas mixture**

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$\therefore 0.662 \times \mathrm{M} = 0.5 \times 0.08 \times 298$

$\Rightarrow \mathrm{M} = 18\ \mathrm{g}$

Let the mole fraction of He in the mixture be a.

So, $\mathrm{XHe_{He}} = \mathrm{a}, \mathrm{XO_2} = (1-a)$

Average molar mass

$M = a \times M_{\mathrm{He}} + (1-a)M_{\mathrm{O_2}}$

$\Rightarrow 18 = a \times 4 + (1-a)32$

$\Rightarrow a = \dfrac{14}{28} = 0.5$

$\therefore 1 - a = 0.5$

$$\%\text{ by mass of He} = \frac{\text{mass of helium}}{\text{Total mass of gases}} \times 100$$

$$= \frac{0.5 \times 4}{0.5 \times 4 + 0.5 \times 32} \times 100$$

$$= 11.11\%$$

Q.8 Answer:

$140\ \mathrm{mm}$ of Hg

Solution:

$\mathrm{p_T} = 84\ \mathrm{mm}, \mathrm{p_A^\circ} = 70\ \mathrm{mm}, \mathrm{X_A} = 0.8, \mathrm{X_B} = 0.2$

$\mathrm{p_T} = 84\ \mathrm{mm}, \mathrm{p_A^\circ} = 70$

$$\mathrm{p_B^\circ} = \frac{\mathrm{p_T} - \mathrm{p_A^\circ} \times \mathrm{X_A}}{\mathrm{X_B}}$$

Q.9 Answer:

For a binary solution AB, if $\mathrm{F_{A-B}} > \mathrm{F_{A-A}}$ and $\mathrm{F_{B-B}}$, there is negative deviation from ideality or Rault's law.

Solution:

Stronger resultant molecular attraction emerges

$\Rightarrow$ vapour pressure less $\Rightarrow$ negative deviation.

Weaker resultant force $\Rightarrow$ positive deviation.

Solution formation is spontaneous even in those cases where $\Delta\mathrm{H} = 0$ or $\Delta\mathrm{H} = +$ ve, because entropy changes, $\Delta\mathrm{S}$ is always +ve.

Q.10 Answer:

the vapour pressure of solution $(\mathrm{n_A} = \mathrm{n_B})$ is less than the pure B component.

Solution:

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$$p = 110x_A + 125x_B$$
$$n_A = n_B$$
$$\therefore \quad x_A = x_B = \frac{1}{2}$$
$$p = 110 \times \frac{1}{2} + 125 \times \frac{1}{2} = 117.5 \text{ mmHg}$$

When B is pure $x_B = 1$ and $x_A = 0$

$p_B^\circ = 125$ mmHg

Hence, $p < p_B^\circ$, 1 st statement

When A is pure $x_A = 1, x_B = 0$

$$p_A^\circ = 110$$

Hence, $p > p_A^\circ$, 2nd statement.

Q.11 Answer:

The composition of the original liquid was $x_A = 0.6$

Solution:

$x_A'' P_A^\circ + x_B'' P_B^\circ = 0.30 P_A^\circ + 0.70 P_B^\circ = 600 \ldots\ldots\ldots\ldots (2)$ **If moles of A and B initially are x and y then**

$$x_A P_A^\circ + x_B P_B^\circ = 700$$
$$x_A'' P_A^\circ + x_B'' P_B^\circ = 0.30 P_A^\circ + 0.70 P_B^\circ = 600$$

$x = 0.75 \times \frac{2}{3}(x+y) + 0.30 \times \frac{1}{3}(x+y)$ **and** $x_A = \frac{x}{x+y}$ **or** $x_B = \frac{x}{x+y}$

Solving gives $x_A = 0.6, x_B = 0.4, P_A^\circ = \frac{2500}{3}$ **torr and** $P_B^\circ = 500$ **torr**

Q.12 Answer:

110

Solution:

(i) Pure liquid component A, $x_A = 1$ and $x_B = 0$

Thus, $p = 110 \times 1 + 125 \times 0 = 110$

Similarly, for pure B, $x_B = 1; x_A = 0, p_B^\circ = 125$

(ii) $x_A = 0.3$ **(given), then** $x_A = 0.7$

$$p = p_A^\circ x_A + p_B^\circ (1 - x_A) = 110 \times 0.3 + 12.5 \times 0.7 = 120.5$$

$$y_A = 33$$

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$y_A = \frac{p_A^x x_A}{p} = \frac{33}{120.5} = 0.27$ **and** $y_B = \frac{p_B^x x_B}{p} = \frac{87.5}{120.5} = 0.73$
(iii) $p_A^\circ = 110, p_B^\circ = 125$

$$p = p_A^\circ x_A + p_B^\circ x_B = 120.5$$

$p_A{}^\circ < p < p_B{}^\circ$
$(15)(C), (16)(C), (17)(C)$
(i) $100X_A + 400\,(1 - X_A) = 380 \Rightarrow 20 = 300X_A$

Q.13 Answer:

0.73

Solution:

(i) Pure liquid component A, $x_A = 1$ **and** $x_B = 0$
Thus, $p = 110 \times 1 + 125 \times 0 = 110$
Similarly, for pure $B, x_B = 1; x_A = 0, p_B{}^\circ = 125$
(ii) $x_A = 0.3$ **(given), then** $x_A = 0.7$

$$p = p_A{}^\circ x_A + p_B{}^\circ\,(1 - x_A) = 110 \times 0.3 + 12.5 \times 0.7 = 120.5$$

$$y_A = 33$$

$y_A = \frac{p_A^x x_A}{p} = \frac{33}{120.5} = 0.27$ **and** $y_B = \frac{p_B^x x_B}{p} = \frac{87.5}{120.5} = 0.73$
(iii) $p_A^\circ = 110, p_B^\circ = 125$

$$p = p_A^\circ x_A + p_B^\circ x_B = 120.5$$

$p_A{}^\circ < p < p_B{}^\circ$
$(15)(C), (16)(C), (17)(C)$
(i) $100X_A + 400\,(1 - X_A) = 380 \Rightarrow 20 = 300X_A$

Q.14 Answer:

$p > p_A{}^\circ$

Solution:

(i) Pure liquid component A, $x_A = 1$ **and** $x_B = 0$
Thus, $p = 110 \times 1 + 125 \times 0 = 110$
Similarly, for pure $B, x_B = 1; x_A = 0, p_B{}^\circ = 125$
(ii) $x_A = 0.3$ **(given), then** $x_A = 0.7$

$$p = p_A{}^\circ x_A + p_B{}^\circ\,(1 - x_A) = 110 \times 0.3 + 12.5 \times 0.7 = 120.5$$

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$$y_A = 33$$

$y_A = \frac{p_A^x x_A}{p} = \frac{33}{120.5} = 0.27$ **and** $y_B = \frac{p_B^x x_B}{p} = \frac{87.5}{120.5} = 0.73$

(iii) $p_A^\circ = 110, p_B^\circ = 125$

$$p = p_A^\circ x_A + p_B^\circ x_B = 120.5$$

$p_A{}^\circ < p < p_B{}^\circ$

Q.15 Answer:

$1/15$

Solution:

(i) $100X_A + 400\,(1 - X_A) = 380 \Rightarrow 20 = 300X_A$

$$X_A = \frac{1}{15}$$

Q.16 Answer:

$6/7$

Solution:

(i) $100X_A + 400\,(1 - X_A) = 380 \Rightarrow 20 = 300X_A$

$$X_A = \frac{1}{15}$$

Q.17 Answer:

Mole fraction of A in vapour phase decreases

Solution:

(i) $100X_A + 400\,(1 - X_A) = 380 \Rightarrow 20 = 300X_A$

$$X_A = \frac{1}{15}$$

Q.18 Answer:

A-p,q,s; B-p,r ; C- p ; D-p,q,s;

Solution:

((A)-pqs, (B) - pr, (C) - p, (D) - pqs)

(A) $C_2H_5OH - H_2O$ **non-ideal solution +ve deviation.**

(B) $H_2O - HNO_3$ **non-ideal solution -ve deviation.**

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(C) $CH_3OH - CH_3CH_2OH$ **ideal solution.**

(D) $CCl_4 - CH_3OH$ **non-ideal solution +ve deviation.**

Q.19 Answer:

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Solution:

Meq. of acid in $25\ mL = 38.5 \times 1$

$\therefore$ **Meq. of acid in** $1000\ mL = \frac{38.5 \times 1 \times 1000}{25} = 1540$

$\therefore$ **Normality of acid** $= \frac{1540}{1000} = 1.54$

$\therefore$ **Molarity of acid** $= \frac{1.54}{n}$ **(where, n is basicity of acid)**

Also, molality $= \frac{\Delta T}{K_f} = \frac{0.4}{0.52}$ **= molarity (given)**

$$\therefore \frac{0.4}{0.52} = \frac{1.54}{n}$$

$$\therefore n = 2$$

Thus, the acid is dibasic.

Q.20 Answer:

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Solution:

In benzene, $\Delta T_f = K_f m$

$$1.28 = 5.12 \times m$$

$$\therefore m = 0.25$$

In H_2O**, ionization takes place,**

$$\therefore \Delta T_f = iK_f m$$

$$1.40 = i \times 1.86 \times 0.25$$

$$\therefore i = 3$$

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